PART 1 — Data Understanding: Datasets, Primary Keys, Foreign Keys and Relationships

How many datasets were provided?

> **Answer: 10 datasets (CSV files) were provided. 9 are business datasets and 1 is a data dictionary.**

The data dictionary only describes column names, so the analysis itself uses the 9 business datasets.

What does each dataset represent?

Dataset	What it represents
Customers	The people who buy from ShopSmart: name, gender, age, contact details, city and state, registration date and membership segment (Bronze, Silver or Gold).
Orders	Every sale made: who bought it (customer), what was bought (product), where (store), when, how many, price, discount, shipping cost, payment method, sales channel and order status (Completed, Pending or Cancelled).
Order_Items	The line detail of each order: quantity, price, discount and line total. Each order has exactly one line here, so it mirrors the Orders data.
Products	The product catalogue: product name, category, sub-category, brand, cost price and selling price.
Stores	ShopSmart's store locations, with region, state, city and opening date.
Suppliers	The companies that supply products: supplier name, email, phone, city and country.
Inventory	The stock held for each product in a store: stock on hand, reorder level and last restock date.
Returns	Items customers sent back: which order, the return date, the reason and the refund amount.
Employees	Store staff: name, role, the store they work in, hire date and monthly salary.
Data_Dictionary	A reference list that describes the columns of some datasets. It covers only 5 of the 9 business datasets.

Identify the Primary Key of each table

Dataset	Primary key	Additional notes
Customers	**Customer_ID**	Every customer has one unique ID.
Orders	**Order_ID**	Every order has one unique ID.
Order_Items	**Order_Item_ID**	Every order line has one unique ID.
Products	**Product_ID**	Every product has one unique ID.

Dataset	Primary key	Additional notes
Stores	**Store_ID**	Every store has one unique ID.
Suppliers	**Supplier_ID**	Every supplier has one unique ID.
Inventory	**Inventory_ID**	Every stock record has one unique ID.
Returns	**Return_ID**	Every return has one unique ID.
Employees	**Employee_ID**	Every employee has one unique ID.

Identify possible Foreign Keys

#	Table (child)	Foreign key	Points to (parent)	Additional notes
1	**Orders**	**Customer_ID**	Customers (Customer_ID)	Shows which customer placed the order.
2	**Orders**	**Product_ID**	Products (Product_ID)	Shows which product was ordered.
3	**Orders**	**Store_ID**	Stores (Store_ID)	Shows which store made the sale.
4	**Order_Items**	**Order_ID**	Orders (Order_ID)	Links each order line to its order.
5	**Order_Items**	**Product_ID**	Products (Product_ID)	Shows which product is on the line.
6	**Returns**	**Order_ID**	Orders (Order_ID)	Shows which order was returned.
7	**Inventory**	**Product_ID**	Products (Product_ID)	Shows which product the stock record is for.
8	**Inventory**	**Store_ID**	Stores (Store_ID)	Shows which store holds the stock.
9	**Employees**	**Store_ID**	Stores (Store_ID)	Shows which store the employee works in.
10	**Products**	**Brand**	Suppliers (Supplier_Name)	Implied link: Supplier_Name is the brand plus " Supplies". Products has no Supplier_ID column.

5. Draw the relationship between the tables

